



Library Management System

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Key Takeaways

- ◆ Developed the system using .NET Core and Angular, implementing MVC and Clean Architecture principles.
- ◆ Implemented authentication and authorization using JWT token-based security.
- ◆ Integrated the frontend and backend, establishing a robust connection with SQL Server.
- ◆ Utilized Dependency Injection (DI) for better maintainability and scalability.
- ◆ Learn about the Debugging and followed the best coding practices for maintainability

Problem Statement

Developed a full-stack Library Management System using .NET Core and Angular, following Clean Architecture principles. The system includes secure JWT-based authentication, real-time book availability, borrowing and return features, and role-based access control. An intuitive admin panel allows management of books, users, and borrowing records, ensuring a seamless experience for both users and administrators.

Technical Stack

Backend

.Net Core 8.0

C#

Database-SQL

Fronted

Angular 18.0

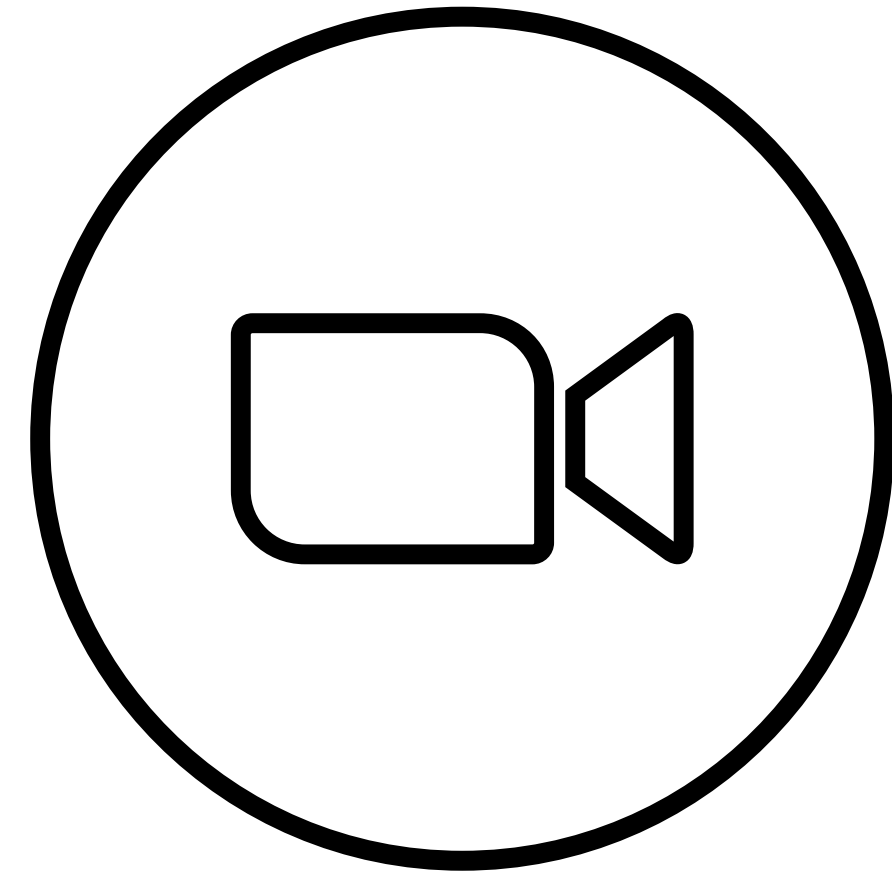
Html/CSS/SCSS

Bootstrap

TypeScript

Angular Material

Project Demo



Images from the Project

The image shows a web application interface for a Library Management System (LMS). The layout includes a dark grey sidebar on the left, a purple header bar at the top, and a purple footer bar at the bottom. The main content area has a dark grey background.

Header: The top purple bar contains the text "Library Management System" on the left, "User" in the center, and a red "Log Out" button on the right.

Sidebar: The left sidebar contains a white box with "LMS" and a list of links: "Books", "Members", and "Loans".

Login Form: A central white box titled "Login Form" contains the following elements:

- An "Email*" input field with the text "pratikaghao@gmail.com".
- A "Password*" input field with masked characters "....." and a toggle icon (an eye with a slash) to its right.
- A green "Login" button.
- A link "Don't have an account?" followed by a green "Register" button.

Footer: The bottom purple bar contains the text "made by: Prathmesh Aghao" on the left and a small white box with a copyright symbol "©" on the right.

Images from the Project

The screenshot displays a web application interface for a Library Management System (LMS). The layout includes a dark grey sidebar on the left, a purple header bar at the top, and a purple footer bar at the bottom.

Header Bar: The title "Library Management System" is centered. On the right, it shows "User" and a red "Log Out" button.

Sidebar: Contains the following links: "LMS" (highlighted with a white border), "Books", "Members", and "Loans".

Main Content Area: Features a "Create Account" form with the following fields and elements:

- Title:** "Create Account" in blue text.
- Name*:** A text input field.
- Email*:** A text input field containing "pratikaghao@gmail.com".
- Password*:** A password input field with masked characters "....." and a toggle icon (an eye with a slash) to the right.
- Register:** A wide, light grey button.
- Already have an account? Login:** A link to the login page.

Footer Bar: Contains the text "made by: Prathmesh Aghao" and a copyright symbol (©) in a square box.

Images from the Project

LMS

Books

Members

Loans

Library Management System

User

Log Out

Library Management System

Get Book by ID:

0

Find Book

Add New Book

Title

Author

ISBN

0

0

0

Add Book

Images from the Project

LMS

Books

Members

Loans

Books List

Atomic Habits

Author: James Clear

ISBN: 978-0735211292

Category ID: 1

Published Year: 2018

Copies Available: 10

Update

Delete

The Alchemist

Author: Paulo Coelho

ISBN: 978-0061122415

Category ID: 2

Published Year: 1993

Copies Available: 7

Update

Delete

1984

Author: George Orwell

ISBN: 978-0451524935

Category ID: 3

Published Year: 1949

Copies Available: 5

Update

Delete

Clean Code

Author: Robert C. Martin

ISBN: 978-0132350884

Category ID: 2

Published Year: 2008

Copies Available: 5

Update

Delete

The Pragmatic Programmer

Author: Andrew Hunt, David Thomas

ISBN: 978-0201616224

Category ID: 2

Published Year: 1999

Copies Available: 3

Update

Delete

You Don't Know JS

Author: Kyle Simpson

ISBN: 978-1491904244

Category ID: 3

Published Year: 2015

Copies Available: 7

Update

Delete

Introduction to Algorithms

Author: Thomas H. Cormen

ISBN: 978-0262033848

Category ID: 4

Published Year: 2009

Copies Available: 4

Update

Delete

Design Patterns

Author: Erich Gamma

ISBN: 978-0201633610

Category ID: 5

Published Year: 1994

Copies Available: 6

Update

Delete

Conclusion

- Successfully built a Library Management System using ASP.NET Core Web API and Angular 18.0.0.
- Applied Clean Architecture, RESTful APIs, authentication, and role-based access.
- Gained hands-on experience in frontend-backend integration, SQL database management, and dependency injection.
- Overcame challenges in API communication, data persistence, and secure user authentication.